SAKET MAHESHWARI

Kasganj, Uttar Pradesh, India
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RESEARCH INTERESTS

EEG-based brain-computer interfaces and neural decoding; biomedical signal processing using time-frequency and wavelet representations; self-supervised and deep learning methods for speech and audio intelligibility; EEG-to-speech reconstruction via neural vocoder frameworks. Broadly interested in applying machine learning to neurological and physiological signal analysis, with a focus on low-resource and clinically relevant settings.

EDUCATION

Centre for Development of Advanced Computing (CDAC) Mohali **2026 – Present**
M.Tech in Artificial Intelligence *India*

GLA University **Nov 2022 – May 2026**
Bachelor of Technology in Computer Science (AI & Machine Learning) *Mathura, Uttar Pradesh*

Mount Litera Zee School **2019**
Class 12 – Computer Programming

PUBLICATIONS

[C1] R. Buragohain, **S. Maheshwari** and K. Nathwani, "Exploiting EEG-based Gamma-Band Time Frequency Feature in WaveNet Decoder Framework for High-Fidelity Speech Reconstruction," *Proc. Interspeech 2026*, pp. 4796–4800, 2026, doi: 10.21437/Interspeech.2026-377.

[C2] S. Maheshwari, R. Buragohain and H. Rajoria, 'Exploring Benchmarking YOLO Frameworks for Automated Aircraft Fuselage Hole Detection," *INDISCON 2026*, **Accepted.**

[C3] S. Maheshwari, R. Chatterjee and S. Sinha, "HybridTrust: Fusing LP-GRU and Transformer Architectures for Enhanced Misinformation Detection in Social Networks," *AECTSD 2025*, COIMBATORE, India, pp. 1–6. doi: 10.1109/AECTSD65988.2025.11411161

[C4] S. Maheshwari, R. Chatterjee and M. Agrawal, "DeepNeuroScan: Advanced CNN Architecture for Precise Brain Tumor Identification in Medical Imaging," *IC2E3 2025*, Srinagar Garhwal, India, pp. 1–6. doi: 10.1109/IC2E365635.2025.11167740

[C5] S. Maheshwari, S. Sinha and R. Chatterjee, "Reducing Aircraft Downtime with Predictive Maintenance: A Data-Driven Approach for Airbus A320," *ICETCI 2025*, Hyderabad, India, pp. 1–9. doi: 10.1109/ICETCI67340.2025.11257885

[C6] S. Maheshwari, M. Agrawal and S. Sinha, "Forecasting Thermal Energy Demand: A Multimodal Approach Using Random Forest and SARIMA Models," *IC2E3 2025*, Srinagar Garhwal, India, pp. 1–5. doi: 10.1109/IC2E365635.2025.11167471

[C7] S. Maheshwari, "Improved Diabetes Prediction Through Multimodeling: Merging Diverse Machine Learning Models for Superior Performance," *ICETI4T 2025*, Navi Mumbai, India, pp. 1–7. doi: 10.1109/ICETI4T63625.2025.11132156

[C8] P. K. Jha, **S. Maheshwari**, R. Kumar and B. Kumar, "Enhancing Network Security through Real-Time Anomaly Detection in Network Traffic," *ICoICC 2025*, Bhubaneswar, India, pp. 1–6. doi: 10.1109/ICoICC64033.2025.11051998

[C9] S. Sinha, **S. Maheshwari** et al., "Predicting Telecom Customer Churn with Combined Machine Learning Models," *ETCOM 2025*, Mangalore, India, pp. 1–7. doi: 10.1109/ETCOM66606.2025.11437063

[B1] S. Maheshwari, S. Sinha and S. Upadhyay, "Unlocking Dev Nagri: OCR with CNN," in *Applied Intelligence and Computing*, SCRS, India, 2024, pp. 157–167. doi: 10.56155/978-81-955020-9-7-17

RESEARCH EXPERIENCE

Research Intern – Signal Processing & Speech AI **Nov 2025 – Present**
Indian Institute of Technology Jammu | Advisor: Prof. Karan Nathwani *Jammu & Kashmir, India*

- Investigated EEG-to-speech reconstruction using gamma-band time-frequency features with a WaveNet decoder; work accepted at **Interspeech 2026** — a top-tier venue in speech and language processing
- Designed a self-supervised speech intelligibility prediction system using Whisper encoder embeddings and BiLSTM; achieved RMSE 0.052 and Pearson correlation 0.89 on a large-scale FLAC dataset without manual annotations
- Built end-to-end EEG signal preprocessing pipelines using wavelet transforms and spectral filtering, achieving 92% classification accuracy on biomedical benchmarks
- Co-authored and contributed to 5+ paper submissions in EEG signal processing, speech reconstruction, and audio intelligibility prediction

Machine Learning Researcher **Jun 2025 – Sep 2025**
National Institute of Technology Jalandhar *Punjab, India*

- Developed transfer learning pipelines using ResNet and VGG for image classification, improving accuracy by 15% over baseline; published in IEEE conference proceedings
- Designed a computer vision framework integrated with AutoCAD to predict and optimize building energy consumption, reducing prediction error by 18%

Machine Learning Intern **Jan 2024 – Mar 2024**
Feynn Labs *Guwahati, Assam, India*

- Developed predictive maintenance and thermal energy forecasting models (Random Forest, LSTM, SARIMA) achieving 87% failure prediction accuracy and 23% lower forecasting error vs. traditional baselines

TECHNICAL SKILLS

Languages & Tools: Python, LaTeX
ML/DL Frameworks: TensorFlow, PyTorch, Keras, scikit-learn, HuggingFace Transformers
Signal Processing & Audio: MNE-Python, librosa, PyWavelets, OpenCV, Whisper
Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook

REFERENCES

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Dr. Sayantan Sinha
Technical Trainer, Dept. of CEA
GLA University Mathura
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